

Example Plots

2025-12-08

Load library and wd

```
suppressMessages(library(tidyverse))
library(knitr)
library(ggplot2)
library(titanic)
library(faraway)
library(faraway)
library(GGally)
```

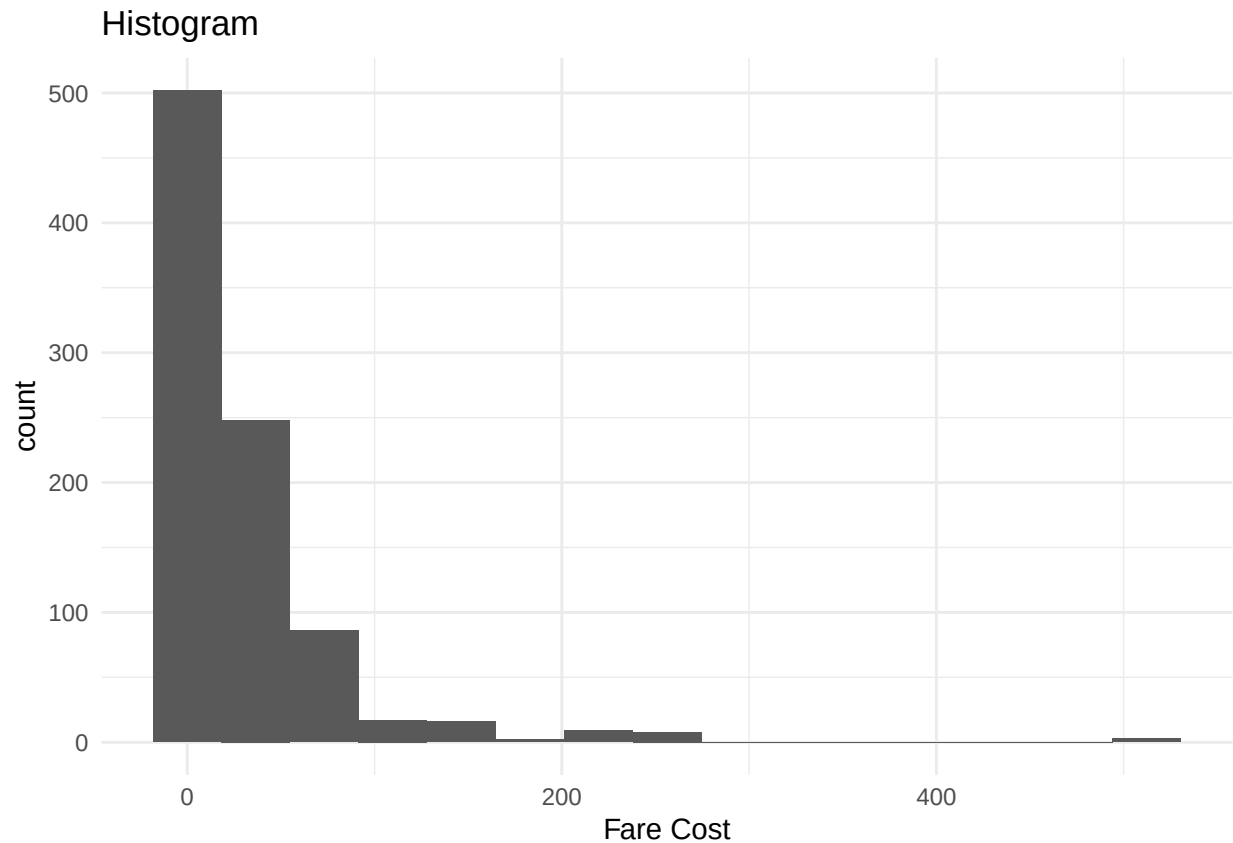
```
##
## Attaching package: 'GGally'

## The following object is masked from 'package:faraway':
##
##     happy
```

Creating Histograms

```
data("titanic_train", package = "titanic")
titanic <- titanic_train
histogram <- ggplot(titanic, aes(x = Fare)) + ## Initialize a ggplot object
  geom_histogram(bins = 15) +
  ggtitle("Histogram") +
  xlab("Fare Cost") +
  theme_minimal()

histogram
```



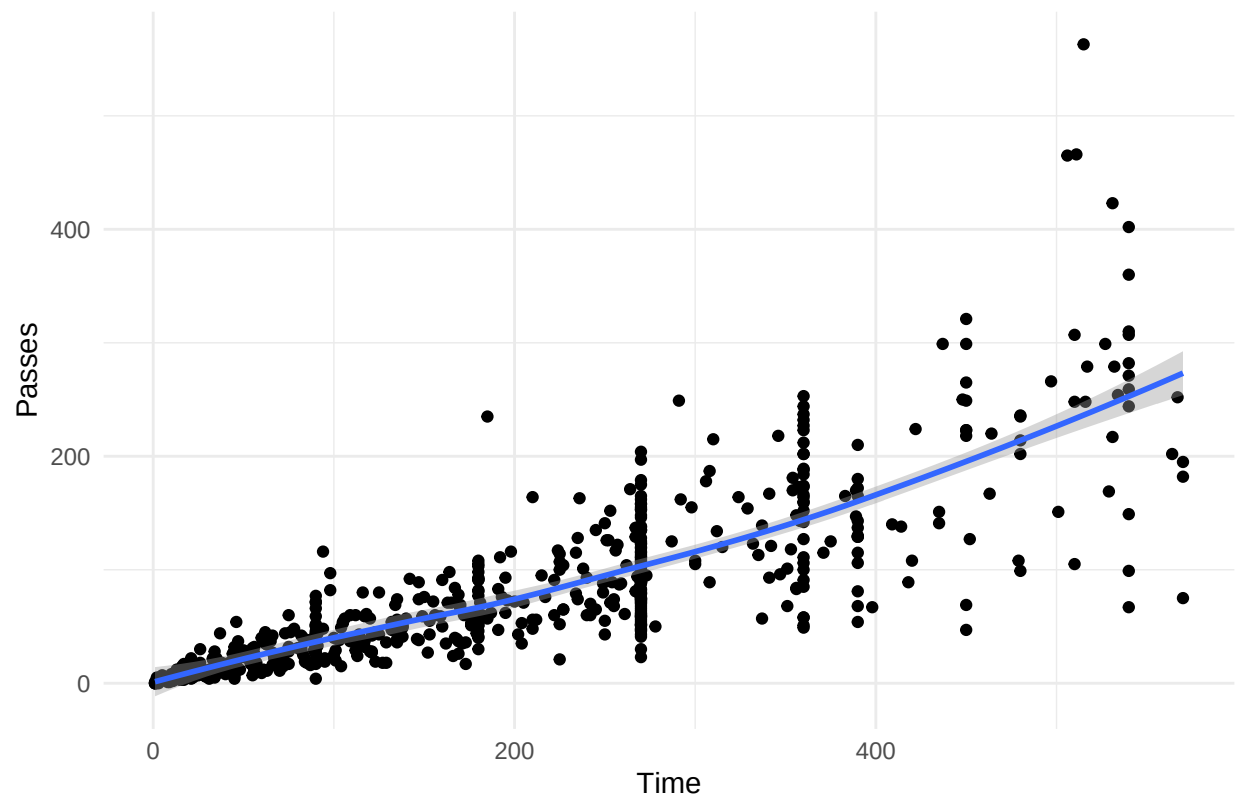
Creating Scatter-plots

```
data("worldcup", package = "faraway")
world_cup <- worldcup
scatter <- ggplot(world_cup, aes(x = Time, y = Passes)) +
  geom_point() +
  geom_smooth() +
  ggtitle("Scatterplot") +
  theme_minimal()

scatter
```

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Scatterplot

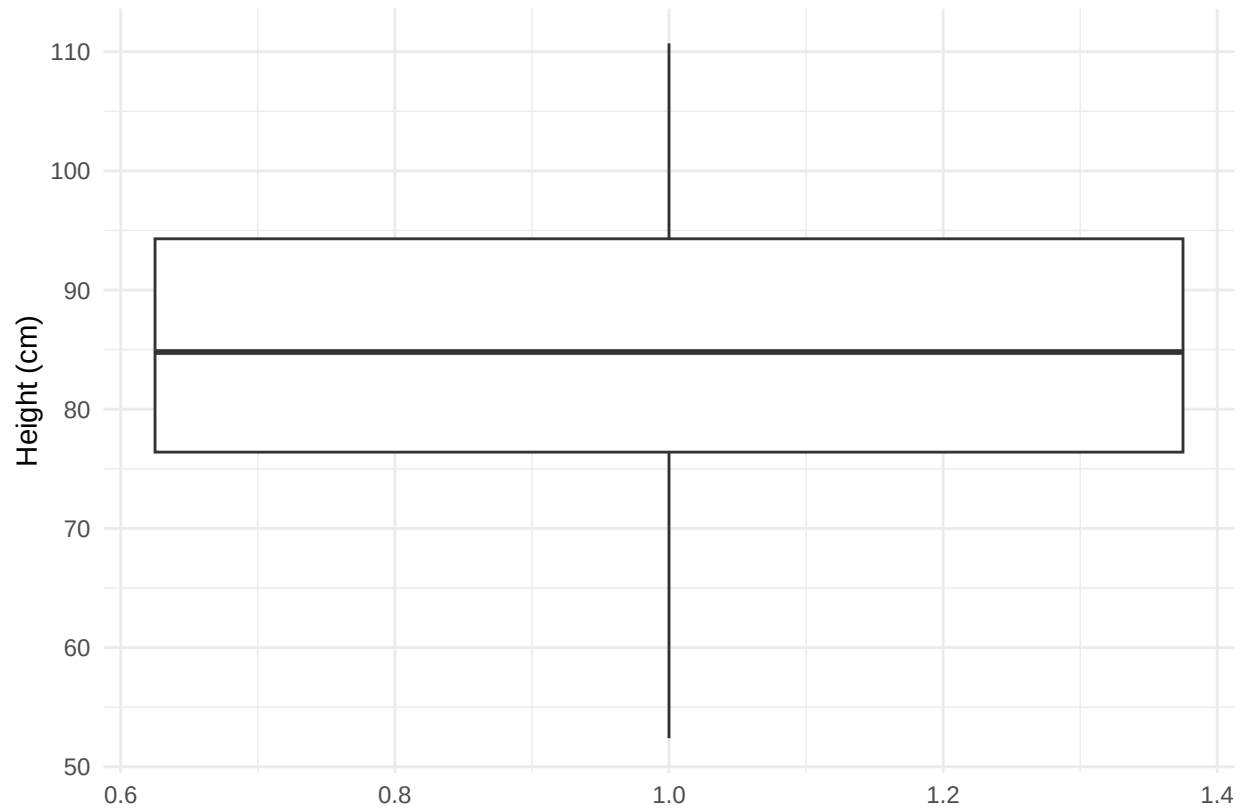


Boxplots

```
boxplot <- ggplot(nepali, aes(x = 1, y = ht)) +  
  geom_boxplot() +  
  xlab("") + ylab("Height (cm)") +  
  theme_minimal()
```

```
boxplot
```

```
## Warning: Removed 123 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```



Exploratory data analysis

```
eda <- ggpairs(titanic %>% select(Fare, Age, Survived))
```

Data Cleaning

```
nepali <- nepali %>%  
  select(id, sex, wt, ht, age) %>%  
  mutate(id = factor(id),  
         sex = factor(sex, levels = c(1, 2),  
                      labels = c("Male", "Female")))) %>%  
  distinct(id, .keep_all = TRUE)
```